

Explore the Power of Octave with Polynomials

This is the tenth article in the mathematical journey through open source and it explains the advanced polynomial mathematics in Octave.



Let's start with the solution to our previous brain teaser of finding the square and cube roots of the imaginary number i , which implies finding all the roots of the equations $x^2 = i$ and $x^3 = i$, respectively. In other words, we need to find roots of the polynomials $x^2 - i$ and $x^3 - i$, respectively, as follows:

```
$ octave -qf
octave:1> roots([1 0 -i])
ans =

    0.70711 + 0.70711i
   -0.70711 - 0.70711i

octave:2> roots([1 0 0 -i])
```

```
ans =

-0.86603 + 0.50000i
-0.00000 - 1.00000i
 0.86603 + 0.50000i
```

```
octave:3>
```

Visualising polynomials

Before we start exploring polynomials further, here are a couple of basic Octave functions to visualise polynomials. `polyout()` displays a polynomial in the usual known format. `polyreduce()` removes the redundant leading zero coefficients. Check out the following: